

Four-Core Pipelined Implied Volatility & Greeks Engine (Artix-7 200T @ 100 MHz)

AXI4-Stream Ingress
(256-bit S, K, T, r, C, TID)

Stage 1: Ingress Registration & TID Queuing (1 Cycle)

- AXI4-Stream 256-bit Ingress {S, K, T, r, C_mkt, TID}
- Direct Ingress Registration
- Zero Serial Ingress Divide
- Single-Cycle Latency

Zero-BRAM Context Store

- 64 x 32-bit Distributed LUTRAM
- SRL32 Delay Shift Registers
- Zero RAMB36/18 Blocks Used
 - Preserves All BRAM for 10GbE / PCIe Host Subsystems

Context

Contract

Stage 2: Analytical Seeder (Brenner-Subrahmanyam, 64 Cyc)

- $\sigma_0 = \frac{2.5066}{\sqrt{T}} \cdot \frac{C_{mkt}}{(S+K)/2}$
- Shared Digit-Recurrence sqrt(T) (29 Cyc)
 - Non-Restoring Divider (33 Cyc)
- Forwards sqrt(T) Directly to Main Datapath
- Yields $|\sigma_0 - \sigma_{n+1}| \leq 0.08$

$\sigma_0, \sqrt{T}$

Stage 3: Core BS Datapath (126 Cycles, II = 1)

- Scale-Invariant Padé In(S/K) (33 Cyc)
- Matched 33-Cyc CORDIC Log Fallback
 - Horner 5th-Order Poly CDF N(d)
- Fully Unrolled 126-Stage Pipeline
- Single-Slot II=1 Loopback Ingestion

C_{BS}, ν
(126 Cyc)

Stage 4: NR Update & Dividers (33 Cycles)

- Convergence: $|C_{BS} - C_{mkt}| \leq \0.01
 - Pipelined NR Divider: err / Vega
- Parallel Gamma Divider: $\phi / (S \cdot \sigma \cdot \sqrt{T})$
 - Delta: N(d1), Vega: $S \cdot \sigma \cdot \sqrt{T} \cdot \phi$
 - Priority Loopback Arbitration

Converged (97.6%)

Stage 5: Multi-Core Arbiter & 128-Bit Egress (2 Cycles)

- Round-Robin 4-Core Complete Drain
- Packed 128-Bit Single-Flit Word: [127:0] = {TID[5:0], Gamma[25:0], Vega[31:0], Delta[31:0], sigma[31:0]}
- Zero Pipeline Bubble Egress

AXI4-Stream Egress
(128-bit σ + Greeks)

Loopback
(2.4%)
 σ_{n+1}